Drivers of Forest Loss Across Countries

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1 Introduction

Forest loss is an important ecological outcome because it reflects long-run damage to biodiversity, carbon storage, and land systems. In this report, we ask a descriptive question: *what is the relationship between industrial activity and forest loss across countries, and how does that relationship vary when we account for economic development?* Our goal is not to claim that industrial activity *causes* forest loss. Instead, we use regression as a descriptive tool to summarize the shape, direction, and relative strength of these relationships across countries.

This question is valuable for two reasons. First, if industrial activity and forest loss move together in systematic ways across countries, that pattern is itself informative for environmental policy and comparative analysis. Second, economic development may change how industrialization relates to ecological damage. This idea is closely related to the Environmental Kuznets Curve (EKC), which suggests that environmental degradation may rise during early industrialization and later flatten or decline as countries become wealthier and invest in cleaner technology, regulation, and sustainability.

We study this question using the Yale Environmental Performance Index (EPI), a cross-country dataset that combines measures of ecosystem vitality, environmental health, and climate change.

2 Data Source

The data is from the Yale Environmental Performance Index (EPI), which contains environmental and development-related indicators for 220 countries from 1995-2022. The dataset is structured in wide format, with countries as rows and year-specific variables as columns.

We study the relationship between industrial activity, economic development, and ecological damage using a linear modeling framework. Because the data are panel-like but relatively small within each year (220 countries), we evaluate generalizability using a train–test split across years rather than within-year splits. Although this introduces temporal dependence, it preserves sample size, and temporal dependence can be somewhat mitigated by choosing years sufficiently far apart.

3 Operationalization

For this project, the most important practical challenge was missing data. Many variables of interest were unavailable for a substantial number of countries, making data completeness a key criterion for selection. Our first task was to choose appropriate years for training and testing. We chose 2012 as the training year and 2021 as the test year due to their relatively strong coverage, with the 10-year gap helping to partially mitigate temporal dependence inherent in this train–test split approach.

We focus on three main conceptual components of the research question: ecological damage (outcome), industrial activity (primary input), and economic development (contextual variation). These concepts are operationalized using variables from the EPI dataset, as summarized in Table 1, which also reports missingness for the candidate variables in the training year.

Table 1: Candidate variables and missing rows in the training year

Category	Variable	Description	Missing rows
Industrial activity	CDO	CO_2 emissions	10
Industrial activity	SO2	Sulfur dioxide emissions	12
Economic development	GDP	Gross domestic product	22
Economic development	HPE	Anthropogenic PM2.5 exposure	22
Economic development	SNM	Sustainable Nitrogen Management Index	26
Other	POP	Population	10
Ecological damage	FCL	Forest cover loss	70
Ecological damage	FSS	Fish stock status	102
Ecological damage	PFL	Primary forest loss	137

Two ecological damage variables, fish stock status (FSS) and primary forest loss (PFL), were excluded due to high missingness. In addition, FSS poses a comparability issue because it is not meaningful for inland countries. As a result, we focus on FCL as the primary outcome variable. FCL captures both the extent and permanence of deforestation.

Industrial activity is measured primarily using CO_2 emissions (CDO), with SO_2 emissions as an alternative indicator.

To capture economic development, we include GDP, population (POP), anthropogenic PM2.5 exposure (HPE), and sustainable nitrogen management (SNM). GDP serves as a standard measure of economic scale and development, while POP controls for country size and helps separate scale effects from other predictors. HPE reflects average pollution exposure and acts as a proxy for industrial cleanliness and infrastructure maturity. SNM was included to capture agricultural and infrastructure efficiency.

4 Exploratory Visualization and Transformations

Before fitting models, we examined the distributions of the candidate variables. Plots of untransformed as well as log-transformed variables are shown in the Appendix (Figures 3 and 4). Many of the raw variables were strongly right-skewed because countries vary enormously in size, industrialization, and pollution. In the raw scale, a small number of countries stretched the distributions and made the underlying patterns harder to interpret. emissions are a good example: most countries cluster at lower values, while a few very high-emission countries create a long right tail.

To reduce impact of outliers on models and improve accuracy, we applied log transformations to variables that clearly benefited from compression of the upper tail. These included CDO, SO2, GDP, POP, and FCL. In contrast, we left HPE and SNM in their raw scales because they were already reasonably distributed and did not gain interpretability from logging.

We also used pairwise visualizations and a correlation matrix to check for multicollinearity before modeling. The strongest pattern was that log(CDO), log(SO2), log(GDP), and log(POP) were all positively correlated, suggesting substantial overlap in what they measure. By contrast, HPE and SNM were less collinear with the main emissions measures, so they appeared more likely to contribute complementary information.

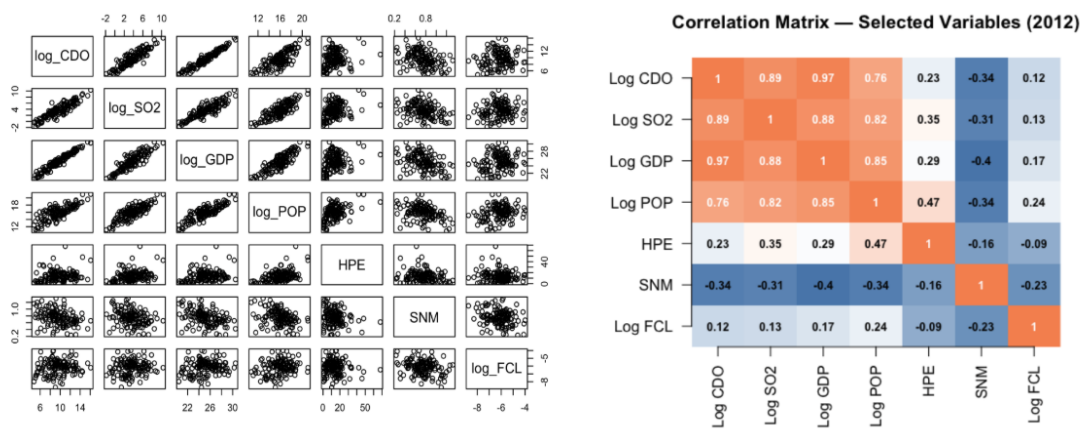

Figure 1: Pairwise scatterplots and correlation matrix for selected predictors in 2012.

These exploratory visuals directly informed the final variable selection. We kept CO_2 emissions as the primary industrial activity measure and dropped SO2 from the final models because it was highly correlated with CO_2 while contributing less distinct signal. We retained GDP, POP, HPE, and SNM because they represented different parts of the broader development and environmental-management story.

5 Modeling Strategy

Following the assignment prompt, we began with the simplest credible model and then built more informative specifications by adding variables that sharpened the descriptive story. The outcome

in every model is log(FCL). Our models were:

$$
\begin{aligned}
\text{Model 1:}\ & \log(\text{FCL}) \sim \log(\text{CDO}) + \log(\text{POP}) \\
\text{Model 2a:}\ & \log(\text{FCL}) \sim \log(\text{CDO}) + \log(\text{GDP}) + \log(\text{POP}) \\
\text{Model 2b:}\ & \log(\text{FCL}) \sim \log(\text{CDO}) + \log(\text{POP}) + \text{HPE} \\
\text{Model 2c:}\ & \log(\text{FCL}) \sim \log(\text{CDO}) + \log(\text{POP}) + \text{HPE} + \text{SNM}
\end{aligned}
$$

Model 1 is the simplest credible model because it pairs a core industrial activity variable with a control for country scale. Model 2a adds GDP as the most obvious economic development measure, but it also serves as a test of what happens when a theoretically relevant variable is nearly collinear with emissions. Model 2b replaces GDP with HPE, which is less collinear and may capture development from a different angle. Model 2c then adds SNM to bring in an agricultural-management dimension that is plausibly relevant to forest loss.

Table 2: Summary of model progression

Model	Specification	R^2	Main takeaway
1	$\log(FCL) \sim \log(CDO) + \log(POP)$	0.064	Weak but interpretable baseline; population significant, CO_2 not significant on its own.
2a	$\log(FCL) \sim \log(CDO) + \log(GDP) + \log(POP)$	0.077	Little improvement; GDP adds limited unique signal because of strong collinearity with CO_2.
2b	$\log(FCL) \sim \log(CDO) + \log(POP) + HPE$	0.129	Clear improvement; HPE contributes new information and is statistically meaningful.
2c	$\log(FCL) \sim \log(CDO) + \log(POP) + HPE + SNM$	0.162	Best-performing descriptive model; all four predictors are statistically significant.

6 Results and Interpretation

Model 1 provided an honest baseline. Population was statistically significant, while CO_2 emissions were not. The negative CO_2 coefficient was directionally consistent with the idea that more developed countries may have lower marginal forest loss, but the signal was weak. This model was useful mainly because it set a low-complexity benchmark.

Model 2a added GDP as a direct measure of economic development, but the gain in fit was minimal. This was not especially surprising: the earlier correlation matrix already showed that GDP and CO_2 emissions were extremely highly correlated. Substantively, Model 2a taught us that a theoretically attractive variable is not always a useful descriptive addition if it largely duplicates information already present in the model.

Model 2b improved substantially. HPE was less collinear with emissions than GDP and entered the model as a meaningful predictor. This suggests that pollution exposure captures a distinct dimension of development and environmental conditions that matters for understanding differences in forest loss across countries.

Model 2c was the strongest model overall. Adding SNM raised R^2 again and made all four predictors statistically significant. The SNM coefficient was especially notable because it was large in magnitude and negative, implying that countries with more sustainable nitrogen management tend to have lower forest cover loss, holding the other variables fixed. In descriptive terms, this suggests that agricultural inefficiency may be closely tied to deforestation patterns, potentially as much as or more than industrial emissions alone.

Taken together, the model sequence supports a clear descriptive story. Emissions and population matter, but a fuller understanding of forest loss across countries comes from adding development and agricultural-management measures that are not simply duplicating the emissions signal.

7 Model Assumptions and Diagnostics

For the final model, we checked standard large-sample regression diagnostics. The residuals-versus-fitted plot did not show a strong systematic pattern, which suggests that the main nonlinear structure had been reasonably captured by the chosen transformations. The Q–Q plot was close to linear with only modest tail deviations, which is acceptable for a cross-country dataset of this size. The Breusch–Pagan test was not statistically significant, so we did not find strong evidence of heteroskedasticity. Finally, the variance inflation factors were low enough to suggest that the remaining predictors were not so collinear as to make the coefficient estimates unstable.

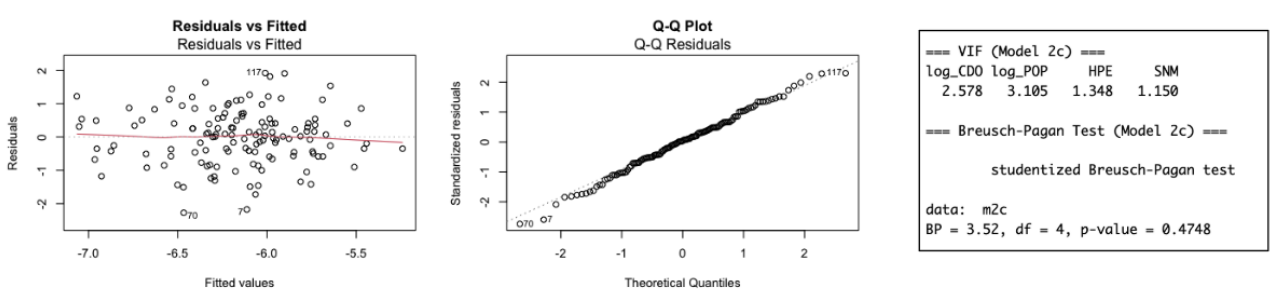

Figure 2: Diagnostic checks for the final model (Model 2c).

These diagnostics do not prove that all classical assumptions hold perfectly, but they do support

the idea that Model 2c is a statistically reasonable descriptive summary of the relationship in the data.

8 Generalizability Across Years

Because this is a descriptive project, we cared not only about in-sample fit but also about whether the model described a stable relationship across time. To assess that, we trained Model 2c on 2012 and then modeled the same variables against 2021 data. The results of the 2021 model are shown in the Appendix (Figure 5). The signs of the coefficients were consistent across years, and the magnitudes were broadly similar. This suggests that the qualitative relationship between industrial activity, development-related variables, and forest loss is fairly stable.

Summary statistics for the 2012 and 2021 data are shown below. While relationships generally held, model performance weakened on the test year: explanatory power (R^2) fell and prediction error (RMSE) rose. That is not a failure so much as a realistic boundary on what a cross-sectional model can capture when applied almost a decade later. Country-level forest dynamics evolve over time as policy, land use, development, and environmental pressures change. The final model therefore appears to capture real and interpretable signal, but not the full complexity of global forest loss.

Table 3: Summary statistics for train and test performance of Model 2c

Metric	Training set (2012)	Test set (2021)
RMSE	0.8253	1.1078
R^2	0.1624	0.0942

9 Conclusion

This report used cross-country EPI data to describe the relationship between industrial activity and forest loss, while allowing that relationship to vary with indicators of development and environmental management. The key descriptive findings are: first, simple industrial activity measures alone provide only a weak summary of forest loss; second, some common development indicators such as GDP are too collinear with emissions to add much unique descriptive value; and third, less collinear variables such as HPE and especially SNM contribute meaningfully to the story.

Our best model, Model 2c, suggests that forest cover loss is associated not only with industrial emissions and country scale, but also with pollution exposure and sustainable nitrogen management. In practical terms, that means the cross-country pattern of forest loss is not purely an emissions story. It is also a story about how development is managed, how pollution is experienced, and how agricultural systems operate.

Appendix

Supplementary Figures

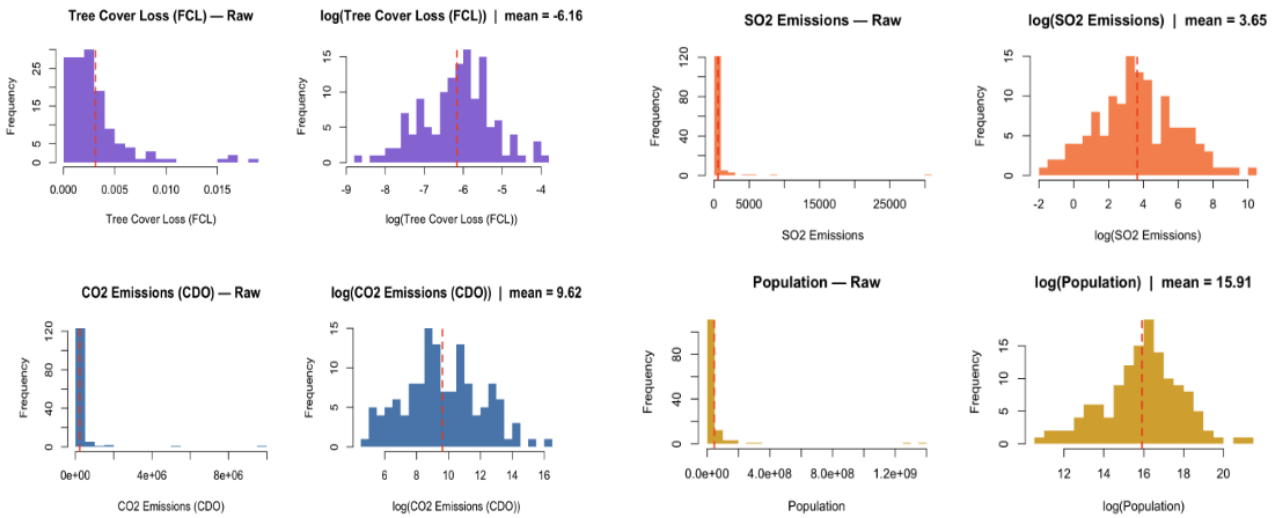

Figure 3: Examples of raw versus log-transformed distributions for variables that benefited from transformation.

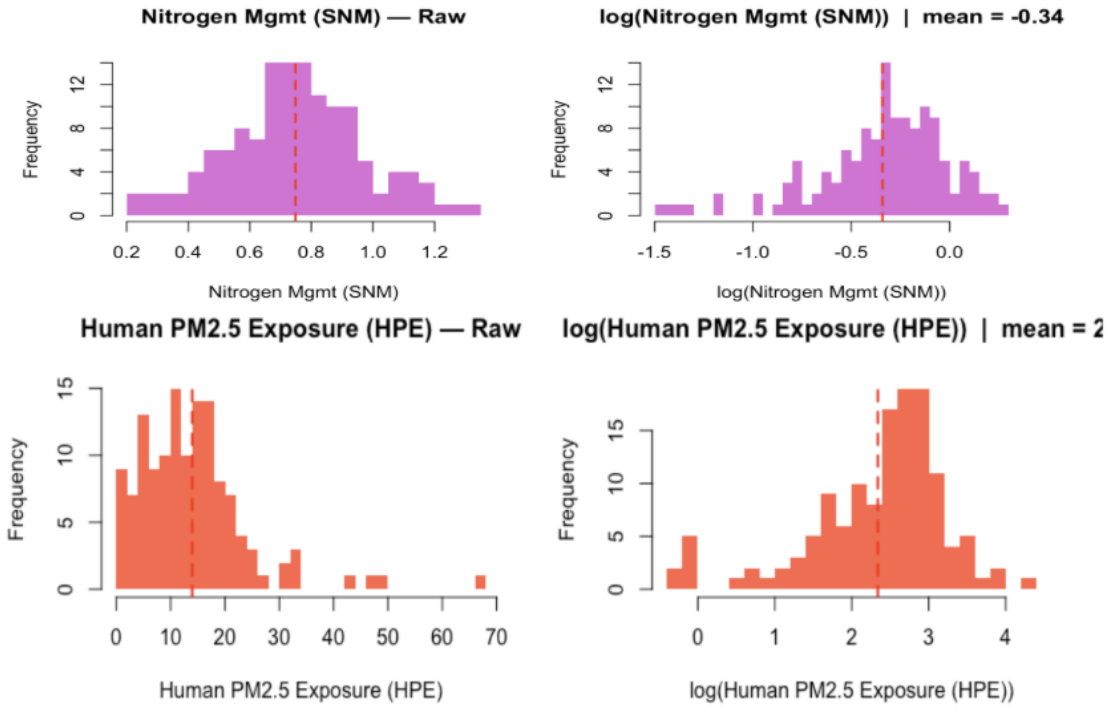

Figure 4: Variables left in raw form because log transformation did not meaningfully improve their distributions.

Model 2c (Training set, 2012)

```
Call:
lm(formula = log_FCL ~ log_CDO + log_POP + HPE + SNM, data = df_train)

Residuals:
     Min       1Q   Median       3Q      Max
-2.27142 -0.49728  0.03221  0.50754  1.91733

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.050695   0.834754  -9.644  < 2e-16 ***
log_CDO     -0.102539   0.048686  -2.106 0.037147 *
log_POP      0.241844   0.066494   3.637 0.000398 ***
HPE         -0.027108   0.008436  -3.213 0.001660 **
SNM         -0.793153   0.349230  -2.271 0.024808 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8413 on 128 degrees of freedom
Multiple R-squared:  0.1624,    Adjusted R-squared:  0.1362
F-statistic: 6.205 on 4 and 128 DF,  p-value: 0.000135
```

Model 2c (Test set, 2021)

```
Call:
lm(formula = log_FCL ~ log_CDO + log_POP + HPE + SNM, data = df_test)

Residuals:
    Min      1Q  Median      3Q     Max
-4.1133 -0.5154  0.0367  0.7132  3.3031

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.89453    1.07645  -8.263 1.61e-13 ***
log_CDO     -0.22636    0.07169  -3.158  0.00199 **
log_POP      0.38822    0.08974   4.326 3.04e-05 ***
HPE         -0.03073    0.01101  -2.792  0.00605 **
SNM         -0.79735    0.43477  -1.834  0.06900 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.088 on 127 degrees of freedom
Multiple R-squared:  0.1595,    Adjusted R-squared:  0.133
F-statistic: 6.024 on 4 and 127 DF,  p-value: 0.00018
```

Figure 5: Diagnostic checks for the final model (Model 2c) on 2012 and 2021 data.

Data Source Link

Yale Environmental Performance Index (EPI): `https://epi.yale.edu/`

Source Code

GitHub repository: https://github.com/qingwong120131/MIDS-203-final-project